

JEE MAIN REVISION MASTER: LAWS OF MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Newton's Laws & Free Body Diagrams	6 PYQs	Dynamics
Moderate	Friction (Static, Limiting & Kinetic)	5 PYQs	Friction

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Newton's Laws & Free Body Diagrams	$\Sigma \vec{F} = m\vec{a}$, $T = \frac{2m_1 m_2}{m_1 + m_2}g$, $a = \frac{m_2 - m_1}{m_1 + m_2}g$
Friction (Static, Limiting & Kinetic)	$f_{\text{max}} = \mu_s N$, $f_k = \mu_k N$, $\theta_{\text{repose}} = \tan^{-1} \mu_s$, $a = g(\sin \theta - \mu_k \cos \theta)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Newton's Laws & Free Body Diagrams (Moderate Yield — 6 PYQs)

Core Principle: Equations of motion for connected bodies, string-pulley constraints, and apparent weight in non-inertial frames.

■ **Caution / JEE Pitfall:** In an accelerating elevator with upward acceleration a , apparent weight is $N = m(g + a)$, NOT $m(g - a)$.

■ **Speed Trick:** Use string constraint $\Sigma T_i \cdot x_i = 0 \implies \Sigma T_i \cdot a_i = 0$ to relate accelerations in complex pulley systems.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string...." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 6m/s^2 . Wha..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "Two masses of 5kg and 3kg are suspended with the help of massless inextensible strings as shown in figure below. When whole system is go..." [Ans: N/A]

3.2 Friction (Static, Limiting & Kinetic) (Moderate Yield — 5 PYQs)

Core Principle: Static self-adjusting friction up to limiting value $\mu_s N$ and constant kinetic friction $\mu_k N$ on rough surfaces.

■ **Caution / JEE Pitfall:** Static friction is self-adjusting: $f_s = F_{\text{(applied)}}$ as long as $F_{\text{(applied)}} \leq \mu_s N$. It does NOT automatically equal $\mu_s N$.

■ **Speed Trick:** In two-block systems, find the maximum common acceleration allowed by friction before determining if slipping occurs.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown i..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "A block of certain mass is placed on a rough floor. The coefficients of static and kinetic friction between the block and the floor are $0.4 \dots$ " **[Ans: N/A]**
- **[2024 Shift] Q#1 (MCQ):** "A block of certain mass is placed on a rough inclined plane. The angle between the plane and the horizontal is 30° . The coefficie..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string.

- (A) $\sqrt{2}Mg$
- (B) $\sqrt{3}Mg$
- (C) Mg
- (D) $3Mg$

Correct Answer: None

Solution Analysis: Let $T_{(1)}$ be the tension in the horizontal string, $T_{(2)}$ be the tension in the slanted string, and $T_{(3)}$ be the tension in the vertical string. The mass M is at rest, so it is in equilibrium. The vertical string supports the weight of the mass. Hence, $T_{(3)} = Mg$. Now consider the knot O . Three tensions act on it: $T_{(1)}$ horizontally, $T_{(2)}$ along the slanted string, and $T_{(3)}$ vertically downward. Since the knot is also in equilibrium, the forces in the vertical direction must balance. $T_{(2)} \sin 45^\circ = T_{(3)} = Mg$. The forces in the horizontal direction must also balance. $T_{(1)} = T_{(2)} \cos 45^\circ = Mg$.

Benchmark Problem 2: KCET 2025 (Q#3)

A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown in the figure. To achieve equilibrium, the co-efficient of static friction between the block on the floor with the floor itself is

- (A) $\mu = \cot \theta$
- (B) $\mu = \sin \theta$
- (C) $\mu = \tan \theta$
- (D) $\mu = \cos \theta$

Correct Answer: None

Solution Analysis: A wooden block of mass M rests on a rough floor, while another block of the same mass is suspended from a point O by strings. To maintain equilibrium, we need to determine the coefficient of static friction between the block on the floor and the floor itself. From the figure and given conditions, we can use the following equations for equilibrium:
The vertical component of tension: $T \sin \theta = Mg \dots (i)$ This equation accounts for the vertical force balance, where T is the tension in the string, and θ is the angle made by the string with the vertical. The horizontal component of